that suggest that there are no answers to questions until they're asked? After all, if the quantum world is contextual, it can't really be like a roulette wheel—there can't be a ball that's on a particular number, passively waiting for us to look, because the properties of the number depend on what we ask about it. The color of a roulette number doesn't change when you ask whether the number is odd; 34 is red, whether or not you ask anything at all about it. Thus, in the quantum world, there can't be a roulette ball until you look for it. As Pascual Jordan said, "We ourselves produce the results of measurement."

Despite the appeal of this Copenhagen-ish argument, Bell handily dismissed it in a "judo-like maneuver" by citing Bohr himself. In the same paper where Bell established contextuality as a key feature of the quantum world, he also pointed out that contextuality shouldn't be a surprise, because, as Bohr said, it's impossible to draw "any sharp distinction between the behaviour of atomic objects and [their] interaction with the measuring instruments." You can't look at the quantum world without altering it—but that doesn't mean the quantum world isn't there before you look. Quite the opposite: if it weren't there, you wouldn't be able to alter it by looking! A contextual roulette wheel can exist—it's just that the ball's location will change when you look at it in different ways, because you can't separate the behavior of the ball from its interaction with you when you look at it. That doesn't mean the ball doesn't exist or that it doesn't have a location before you look; it just means the ball is kind of jumpy and sensitive, moving around dramatically at even the slightest disturbance. The hidden variables in Bohm's pilot-wave interpretation behave in exactly this way. Particles, according to Bohm, always have positions—but those positions can be dramatically altered by small disturbances and changes in experimental setups. Ask a slightly different set of questions to an electron, in Bohm's world, and you can get an enormously different set of answers—but the electron has a definite position all the while. And because Bohm's theory is contextual, it evades all of the proofs that supposedly rule it out. "What is proved by impossibility proofs," concluded Bell, "is lack of imagination."